

Yutian Pang, PhD

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Postdoctoral Fellow

University of Texas at Austin, Department of Aerospace Engineering & Engineering Mechanics

Austin, TX, USA

EDUCATION

- Ph.D., Mechanical and Aerospace Engineering, 08/2018-05/2023
Arizona State University, Tempe, AZ
- M.S., Mechanical and Aerospace Engineering, 08/2017-06/2018
Arizona State University, Tempe, AZ
- B.S., Mechanical Design, Manufacturing, and Automation, 09/2013-06/2017
Huazhong University of Science and Technology, Wuhan, China

PROFESSIONAL EXPERIENCE

- UT Austin, Department of Aerospace Engineering & Engineering Mechanics Austin, TX
Postdoctoral Research Fellow 07/2024 – present
- Thales USA San Jose, CA
Machine Learning Engineer 06/2023 – 07/2024

HONORS AND AWARDS

- | | | |
|------|---|------------|
| 2025 | 2nd Place in FAA ML/AI Data Challenge for Aviation Safety | McLean, VA |
| 2023 | Outstanding Graduate Research Award of ASU MAE | Tempe, AZ |
| 2015 | Outstanding Academic Performance Award of HUST MSE | Wuhan, CN |

JOURNAL PUBLICATIONS

(Pang, Y.* indicates I am the corresponding author)

Co-author relationship key (per department request, for publications in the past five years). Bracketed notes identify my relationship to senior authors and to students I mentored: **Y. Liu** — my Ph.D. advisor; **J. Clarke** — my postdoctoral advisor; students I mentored/advised — **Q. Xu, R. Rathnakumar, J. Liang, A. Porcayo**. All other co-authors are peer collaborators/colleagues.

Articles Under Review

1. **Pang, Y.**, Hu, J., Iyer, H., Zhao, X., Clarke, J. (2026). V-STAR: VectorNet-Augmented Spatio-Temporal Attention and Reasoning for Procedure-Aware Aircraft Trajectory Prediction. Submitted to *Nature Communications*. [J. Clarke: postdoctoral advisor]
2. **Pang, Y.***, Dong, Z., Clarke, J. (2026). Runway Incursion Detection: A Three-Layer Conflict Alert Scheme using ATC Speech and ADS-B Data. Submitted to *Computers in Industry*. [J. Clarke: postdoctoral advisor]
3. **Pang, Y.***, Delahaye, D., Clarke, J. (2026). Trajectory-Based Optimization for Air Traffic Control in the Terminal Maneuvering Area. Submitted to *Transportation Research Part C*. [J. Clarke: postdoctoral advisor]

4. Porcayo, A., **Pang, Y.***, Thomas, M., & Clarke, J. (2026). Machine Learning-based Landing Aircraft Taxiplan Prediction: Case study with KATL. Submitted to *Journal of Air Transport Management*. [A. Porcayo: M.S. student I mentored; J. Clarke: postdoctoral advisor]
5. Liang, J., **Pang, Y.***, & Clarke, J. (2026). Optimizing Same-Day Delivery: A Framework Incorporating Vehicle Capacity and Time Guarantees. Submitted to *Transportation Science*. [J. Liang: Ph.D. student I mentor; J. Clarke: postdoctoral advisor]

Published Journal Articles

1. **Pang, Y.***, Kendall, A., Clarke, J. (2026). Modeling the Impact of Communication and Human Uncertainties on Runway Capacity in Terminal Airspace. *Journal of Air Transport Management*. [J. Clarke: postdoctoral advisor]
2. **Pang, Y.***, Kendall, A., Clarke, J. (2026). The Reliability of Remotely Piloted Aircraft System Performance under Aeronautical Communication Uncertainties. *Reliability Engineering & System Safety*. [J. Clarke: postdoctoral advisor]
3. **Pang, Y.***, Kendall, A. P., Porcayo, A., Barsotti, M., Jain, A., Clarke, J. (2026). From Voice to Safety: Language AI Powered Pilot-ATC Communication Understanding for Airport Surface Movement Collision Risk Assessment. *Transportation Research Part C: Emerging Technologies*, 184, p.105540. [A. Porcayo: M.S. student I mentored; J. Clarke: postdoctoral advisor]
4. Xu, Q., **Pang, Y.**, & Liu, Y. (2025). Data-driven governing equation identification of near terminal air traffic flow dynamics. *Journal of Air Transport Management*. [Q. Xu: Ph.D. student I mentored; Y. Liu: Ph.D. advisor]
5. Xu, Q., **Pang, Y.**, & Liu, Y. (2024). Dynamic airspace sectorization with machine learning enhanced workload prediction and clustering. *Journal of Air Transport Management*. [Q. Xu: Ph.D. student I mentored; Y. Liu: Ph.D. advisor]
6. Xu, Q., **Pang, Y.**, Zhou, X., & Liu, Y. (2024). PIGAT: Physics-Informed Graph Attention Transformer for Air Traffic State Prediction. *IEEE Transactions on Intelligent Transportation Systems*. [Q. Xu: Ph.D. student I mentored; Y. Liu: Ph.D. advisor]
7. **Pang, Y.**, Zhao, P., Hu, J., & Liu, Y. (2024). Machine Learning-Enhanced Aircraft Landing Scheduling under Uncertainties. *Transportation Research Part C: Emerging Technologies*, 158, p.104444. [Y. Liu: Ph.D. advisor]
8. Zhang, L., Wang, M., Ding, Y., Wan, T., Qi, B., **Pang, Y.** (2023). FBC-ANet: A Semantic Segmentation Model for UAV Forest Fire Images Combining Boundary Enhancement and Context Awareness. *Drones*, 7(7), 456. [external collaboration]
9. Rathnakumar, R., **Pang, Y.**, & Liu, Y. (2023). Epistemic and aleatoric uncertainty quantification for crack detection using a Bayesian Boundary Aware Convolutional Network. *Reliability Engineering & System Safety*, 109547. [R. Rathnakumar: Ph.D. student I mentored; Y. Liu: Ph.D. advisor]
10. Hu, J., Xu, Z., Wang, W., Qu, G., **Pang, Y.**, & Liu, Y. (2023). Decentralized graph-based multi-agent reinforcement learning using reward machines. *Neurocomputing*, 126974. [Y. Liu: Ph.D. advisor]
11. Xu, Q., **Pang, Y.**, & Liu, Y. (2023). Air Traffic Density Prediction using Bayesian Ensemble Graph Attention Network (BEGAN). *Transportation Research Part C: Emerging Technologies*. [Q. Xu: Ph.D. student I mentored; Y. Liu: Ph.D. advisor]
12. **Pang, Y.**, Hu, J., Lieber, C., Cooke, N., & Liu, Y. (2023). Air Traffic Controller Cognitive Workload Level Prediction using Conformal Dynamical Graph Learning. *Advanced Engineering Informatics*, 57, p.102113. [N. Cooke, C. Lieber: human-factors collaborators; Y. Liu: Ph.D. advisor]
13. Huang, J., **Pang, Y.**, Zhao, X., Liu, Y., & Yan, H. (2022). Posterior Regularized Bayesian Neural Network Incorporating Soft and Hard Knowledge Constraints. *Knowledge-Based Systems*, 249, p.109047. [H. Yan: IE collaborator; Y. Liu: Ph.D. advisor]

14. Hu, J., Wang, Y., **Pang, Y.**, & Liu, Y. (2022). Optimal maintenance scheduling under uncertainties using Linear Programming-enhanced Reinforcement Learning. *Engineering Applications of Artificial Intelligence*, 109, p.104655. [Y. Liu: Ph.D. advisor]
15. Wei, H., Yao, H., **Pang, Y.**, & Liu, Y. (2022). Fracture Pattern Prediction with Random Microstructure using Physics-Informed Deep Neural Networks. *Engineering Fracture Mechanics*, 108497. [Y. Liu: Ph.D. advisor]
16. **Pang, Y.**, Zhao, X., Hu, J., Yan, H., & Liu, Y. (2022). Bayesian Spatio-Temporal Graph Transformer Network (B-Star) for Multi-Aircraft Trajectory Prediction. *Knowledge-Based Systems*, 249, p.108998. [H. Yan: IE collaborator; Y. Liu: Ph.D. advisor]
17. **Pang, Y.**, Zhao, X., Yan, H., & Liu, Y. (2021). Data-driven trajectory prediction with weather uncertainties: A Bayesian deep learning approach. *Transportation Research Part C: Emerging Technologies*, 130, p.103326. [H. Yan: IE collaborator; Y. Liu: Ph.D. advisor]
18. Wang, Y., **Pang, Y.**, Chen, O., Iyer, H. N., Dutta, P., Menon, P. K., & Liu, Y. (2021). Uncertainty quantification and reduction in aircraft trajectory prediction using Bayesian-Entropy information fusion. *Reliability Engineering & System Safety*, 107650. [Y. Liu: Ph.D. advisor]
19. Wang, Y., **Pang, Y.**, Gorceski, S., Kostiuk, P., Mohen, M. T., Menon, P. K., & Liu, Y. (2021). A Voice Communication-Augmented Simulation Framework for Aircraft Trajectory Simulation. *IEEE Transactions on Intelligent Transportation Systems*. [Y. Liu: Ph.D. advisor]

CONFERENCE PROCEEDINGS & PRESENTATIONS

1. **Pang, Y.**, Delahaye, D., Clarke, J. (2026). Geometric Trajectory Optimization for TRACON Arrivals: An NLP Approach with ATC Vectoring Maneuver Modeling. *Annual Modeling and Simulation Conference 2026*.
2. Momit, M. A., Jiang, W., Hussain, B. Z., Ammar, M., Bhujel, S., **Pang, Y.**, ... & Hu, J. (2026). Deep Q-Network With Lagrangian Relaxation for Autonomous Aircraft Landing. In *AIAA SCITECH 2026 Forum* (p. 1983).
3. Rathnakumar, R., Liu, Y., & **Pang, Y.** (2025). Bayesian Approach for Uncertainty Quantification of Neural Networks-Based Crack Diagnostics. In *AIAA SCITECH 2025 Forum* (p. 1959).
4. Xu, Q., **Pang, Y.**, Zhang, Z., & Liu, Y. (2022). Physics-Based Learning for Aircraft Waiting Time Prediction. In *AIAA AVIATION 2022 Forum*, p.3826.
5. **Pang, Y.**, Xu, N., & Liu, Y. (2022). Robust Satellite Image Classification with Bayesian Deep Learning. In *2022 Integrated Communication, Navigation and Surveillance Conference (ICNS)*.
6. **Pang, Y.**, Cheng, S., Hu, J., & Liu, Y. (2021). Evaluating the Robustness of Bayesian Neural Networks Against Different Types of Attacks. In *CVPR 2021 Workshops on Adversarial Machine Learning in Real-World Computer Vision Systems*. [PDF]
7. **Pang, Y.**, Wang, Y., & Liu, Y. (2020). Probabilistic Aircraft Trajectory Prediction with Weather Uncertainties using Approximate Bayesian Variational Inference to Neural Networks. In *AIAA AVIATION 2020 Forum*, p.2897.
8. **Pang, Y.**, & Liu, Y. (2020). Conditional Generative Adversarial Networks (CGAN) for Aircraft Trajectory Prediction considering weather effects. In *AIAA Scitech 2020 Forum*, p.1853.
9. **Pang, Y.**, & Liu, Y. (2020). Probabilistic Aircraft Trajectory Prediction Considering Weather Uncertainties Using Dropout As Bayesian Approximate Variational Inference. In *AIAA Scitech 2020 Forum*, p.1413.
10. **Pang, Y.**, Hu, J., Cheng, S., & Liu, Y. (2019, September). Aircraft trajectory prediction using LSTM neural network with embedded convolutional layer. In *11th Annual Conference of the Prognostics and Health Management Society, PHM 2019*. Prognostics and Health Management Society.

11. **Pang, Y.**, Yao, H., Hu, J., & Liu, Y. (2019). A Recurrent Neural Network Approach for Aircraft Trajectory Prediction with Weather Features From Sherlock. In *AIAA Aviation 2019 Forum*, p.3413.
12. Wang, Y., **Pang, Y.**, Liu, Y., Dutta, P., & Yang, B. J. (2019). Aircraft Trajectory Prediction and Risk Assessment Using Bayesian Updating. In *AIAA Aviation 2019 Forum*, p.2936.
13. Zhao, X., Yan, H., Li, J., **Pang, Y.**, & Liu, Y. (2019, September). Spatio-temporal Anomaly Detection, Diagnostics, and Prediction of the Air-traffic Trajectory Deviation using Convective Weather. In *Annual Conference of the PHM Society*, Vol. 11.

U.S. PATENTS

1. Yongming Liu, **Yutian Pang**, and Jueming Hu. *Systems and Methods for Machine Learning-Enhanced Aircraft Landing Scheduling Under Uncertainties*.
2. Yongming Liu, **Yutian Pang**, and Jueming Hu. *Systems and Methods for Air Traffic Controller Workload Level Prediction using Conformalized Dynamical Graph Learning*.
3. Yongming Liu, Qihang Xu, **Yutian Pang**. *Systems and Methods for Dynamic airspace sectorization with machine learning enhanced workload prediction and clustering*.

INVITED TALKS

1. INFORMS Annual Conference, *Evaluation of Near-Terminal Operation Performance under Uncertainties in High-Density Airspace*. Innovative Approaches to Efficient Airspace Management, Oct. 2025, Atlanta, GA.
2. INFORMS Annual Conference, *Machine Learning Based Aircraft Trajectory Prediction With Historical Data*. Advances in Data-Driven Air Traffic Flow Management, Oct. 2021, Anaheim, CA.

PROPOSAL EXPERIENCE

(Listing proposals Yutian Pang as the Major Initiator/Writer)

1. NASA NAMS-2 Task. Submitted. \$ 100K.
2. Airport Cooperative Research Program, Transportation Research Board. Not selected. \$ 400K.
3. NASA ULI Round 9, Thrust 5: In-Time System-Wide Safety Assurance. Not selected. \$5M.
4. 2024 FAA Machine Learning/Artificial Intelligence Challenge for Aviation Safety. Awarded. \$20K.

SYNERGISTIC ACTIVITIES

1. Technical Program Committee for CVPR workshop on Perception Beyond the Visible Spectrum, 2024-current.
2. Early Career Editorial Board Member for journal *Computer Systems Science and Engineering*, 2023-current.

MENTORING SERVICES

UT Austin

1. Cameron Jobson, B.S. in Computer Science. Ongoing.
2. Viraj Singhee, B.S. in Computer Science. Ongoing.
3. Jianhua Liang, Ph.D. in Aerospace Engineering. Ongoing. Two top journal submitted.
4. Alex Porcayo, M.S. in Aerospace Engineering, Graduated 2025. Two top journal submitted.

ASU

1. Rahul Rathnakumar, PhD, MAE, Graduated 2024.
2. Qihang Xu, PhD, MAE, Graduated 2024.

PEER REVIEWER

Peer reviewer for 40+ journals in aerospace, mechanical, transportation, AI, IE/OR, and CS/CE, including Aerospace Science and Technology, Chinese Journal of Aeronautics, AIAA Journal of Aerospace Information Systems, Computers in Industry, Information Fusion, Reliability Engineering & System Safety, Transportation Research Part C/D/E/F, IEEE TITS/TAES, etc.



Yutian Pang

PostDoc at UT Austin

Applied AI
Automation
CNS/ATM
Safety & Security
Risk & Reliability

	All	Since 2021
Citations	1089	1060
h-index	19	19
i10-index	23	23

3 articles

19 articles

not available

available

Based on funding mandates

TITLE	CITED BY	YEAR
Data-driven trajectory prediction with weather uncertainties: A Bayesian deep learning approach Y Pang, X Zhao, H Yan, Y Liu Transportation Research Part C: Emerging Technologies 130, 103326	127	2021
Conditional generative adversarial networks (CGAN) for aircraft trajectory prediction considering weather effects Y Pang, Y Liu AIAA Scitech 2020 Forum, 1853	100	2020
Bayesian spatio-temporal graph transformer network (b-star) for multi-aircraft trajectory prediction Y Pang, X Zhao, J Hu, H Yan, Y Liu Knowledge-Based Systems 249, 108998	98	2022
A recurrent neural network approach for aircraft trajectory prediction with weather features from sherlock Y Pang, H Yao, J Hu, Y Liu AIAA Aviation 2019 Forum, 3413	81	2019
Air traffic density prediction using Bayesian ensemble graph attention network (BEGAN) Q Xu, Y Pang, Y Liu Transportation Research Part C: Emerging Technologies 153, 104225	76	2023
Aircraft trajectory prediction using LSTM neural network with embedded convolutional layer Y Pang, N Xu, Y Liu Proceedings of the Annual Conference of the PHM Society 11, 1-8	60	2019
PIGAT: Physics-informed graph attention transformer for air traffic state prediction Q Xu, Y Pang, X Zhou, Y Liu IEEE Transactions on Intelligent Transportation Systems 25 (9), 12561-12577	59	2024
Uncertainty quantification and reduction in aircraft trajectory prediction using Bayesian-Entropy information fusion Y Wang, Y Pang, O Chen, HN Iyer, P Dutta, PK Menon, Y Liu Reliability Engineering & System Safety 212, 107650	59	2021

TITLE	CITED BY	YEAR
Optimal maintenance scheduling under uncertainties using Linear Programming-enhanced Reinforcement Learning J Hu, Y Wang, Y Pang, Y Liu Engineering Applications of Artificial Intelligence 109, 104655	49	2022
Air traffic controller workload level prediction using conformalized dynamical graph learning Y Pang, J Hu, CS Lieber, NJ Cooke, Y Liu Advanced Engineering Informatics 57, 102113	43	2023
Probabilistic aircraft trajectory prediction considering weather uncertainties using dropout as Bayesian approximate variational inference Y Pang, Y Liu AIAA Scitech 2020 Forum, 1413	42	2020
Machine learning-enhanced aircraft landing scheduling under uncertainties Y Pang, P Zhao, J Hu, Y Liu Transportation Research Part C: Emerging Technologies 158, 104444	35	2024
Epistemic and aleatoric uncertainty quantification for crack detection using a Bayesian Boundary Aware Convolutional Network R Rathnakumar, Y Pang, Y Liu Reliability Engineering & System Safety 240, 109547	28	2023
Fracture pattern prediction with random microstructure using a physics-informed deep neural networks H Wei, H Yao, Y Pang, Y Liu Engineering Fracture Mechanics 268, 108497	27	2022
Decentralized graph-based multi-agent reinforcement learning using reward machines J Hu, Z Xu, W Wang, G Qu, Y Pang, Y Liu Neurocomputing 564, 126974	24	2024
FBC-ANet: A semantic segmentation model for UAV forest fire images combining boundary enhancement and context awareness L Zhang, M Wang, Y Ding, T Wan, B Qi, Y Pang Drones 7 (7), 456	23	2023
Evaluating the Robustness of Bayesian Neural Networks Against Different Types of Attacks Y Pang, S Cheng, J Hu, Y Liu CVPR 2021 Workshop on Adversarial Machine Learning in Real-World Computer ...	23	2021
A voice communication-augmented simulation framework for aircraft trajectory simulation Y Wang, Y Pang, S Gorceski, P Kostiuik, MT Mohen, PK Menon, Y Liu IEEE Transactions on Intelligent Transportation Systems 23 (7), 7310-7320	23	2021
Posterior Regularized Bayesian Neural Network incorporating soft and hard knowledge constraints J Huang, Y Pang, Y Liu, H Yan	21	2023

TITLE	CITED BY	YEAR
Knowledge-Based Systems 259, 110043		
Aircraft trajectory prediction and risk assessment using bayesian updating Y Wang, Y Pang, Y Liu, P Dutta, BJ Yang Aiaa aviation 2019 forum, 2936	19	2019